INTERNAL — CONFIDENTIAL

SignalRisk

Dashboard User Guide

A complete guide to the SignalRisk analyst dashboard — case management, fraud operations, rule authoring, and real-time monitoring.

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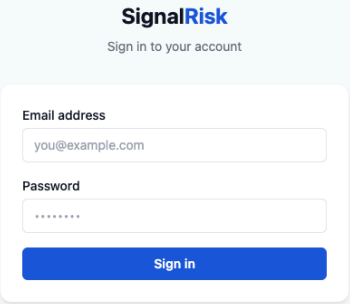
1 Getting Started

Login, session management, and dashboard navigation



Login page — enter email and password to access the dashboard

Logging In

Navigate to your dashboard URL and enter your credentials. The system uses JWT authentication with 15-minute access tokens, automatically renewed while you remain active.

Account Types

- **Admin** — full access including Admin panel
- **Analyst** — Cases, Fraud Ops, Analytics, Live Feed
- **Merchant** — own data only

Development Credentials

- `admin@signalrisk.com` / `password`
- `analyst@signalrisk.com` / `password`

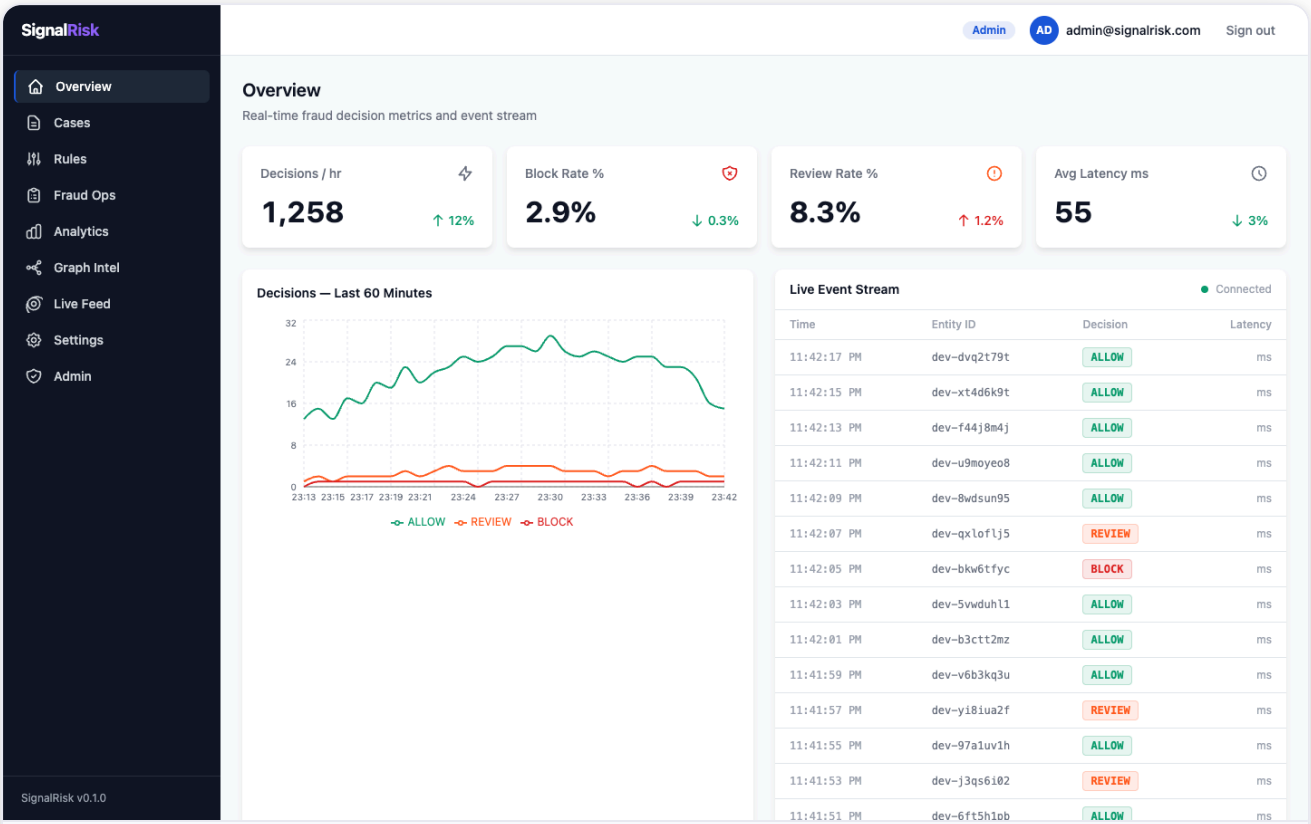
Navigation

The left sidebar links to all pages. Your role badge and email appear in the top-right. Click **Sign out** to revoke your JWT token and end the session.

> **NOTE**
> Navigating to a protected URL while logged out redirects to the login page. After login you are forwarded to the originally requested page.

2 Overview

Real-time KPI metrics, decision trend chart, and live event stream



Overview page — KPI cards (top), 60-minute decision trend chart (left), live event stream (right)

KPI Cards

Four metric cards update automatically every 30 seconds:

Card	Description
Decisions / hr	Fraud decisions made in the last hour
Block Rate %	Percentage of decisions resulting in BLOCK
Review Rate %	Percentage of decisions resulting in REVIEW
Avg Latency ms	Average decision processing time

Decision Trend Chart

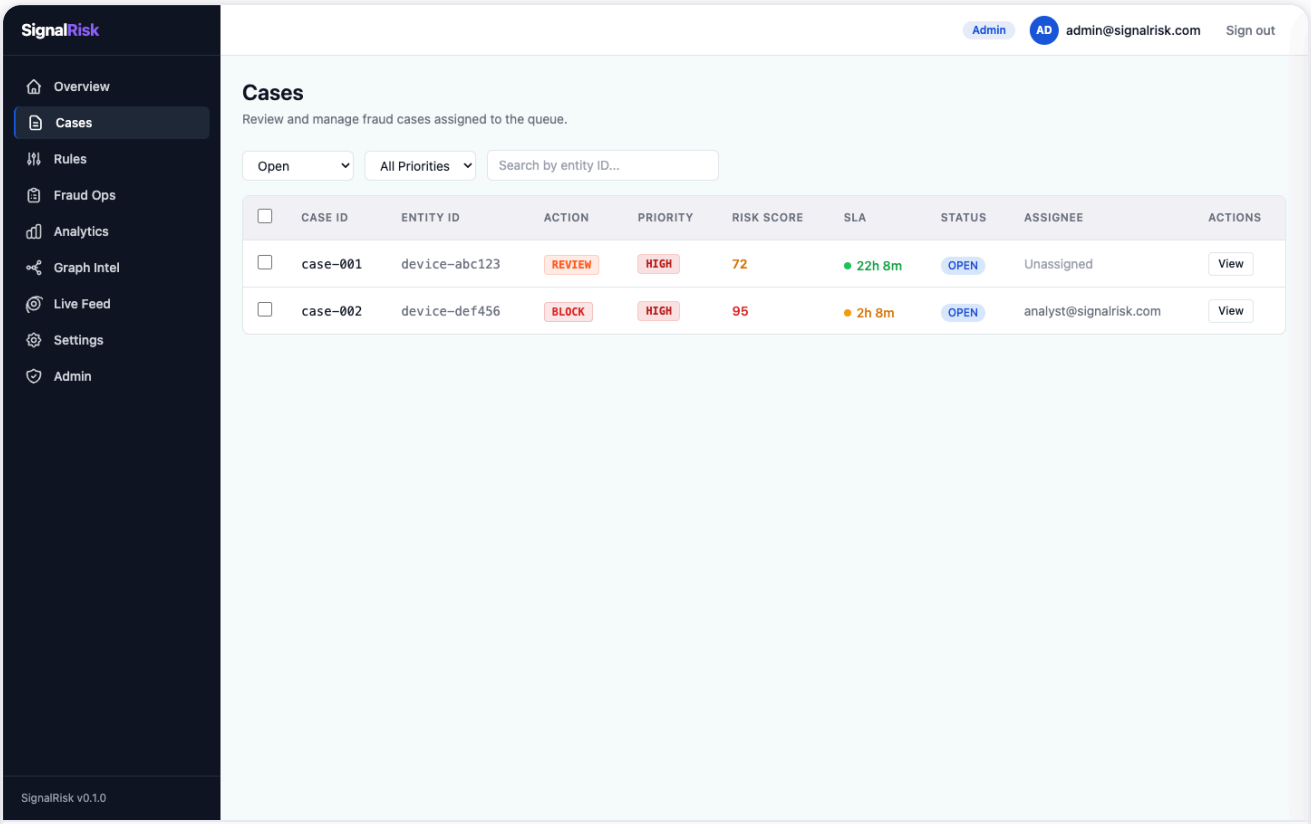
Decision volume by outcome (ALLOW / REVIEW / BLOCK) for the last 60 minutes at 2-minute intervals. Lines: green (ALLOW), amber (REVIEW), red (BLOCK).

Live Event Stream

Right panel shows decisions in real time via WebSocket. Each row: timestamp, entity ID, decision badge, latency. Green **Connected** indicator confirms the WebSocket is active.

3 Cases

Analyst review queue — resolve, escalate, and manage fraud cases



Cases page — OPEN queue with status, priority, risk score, SLA, and assignee columns

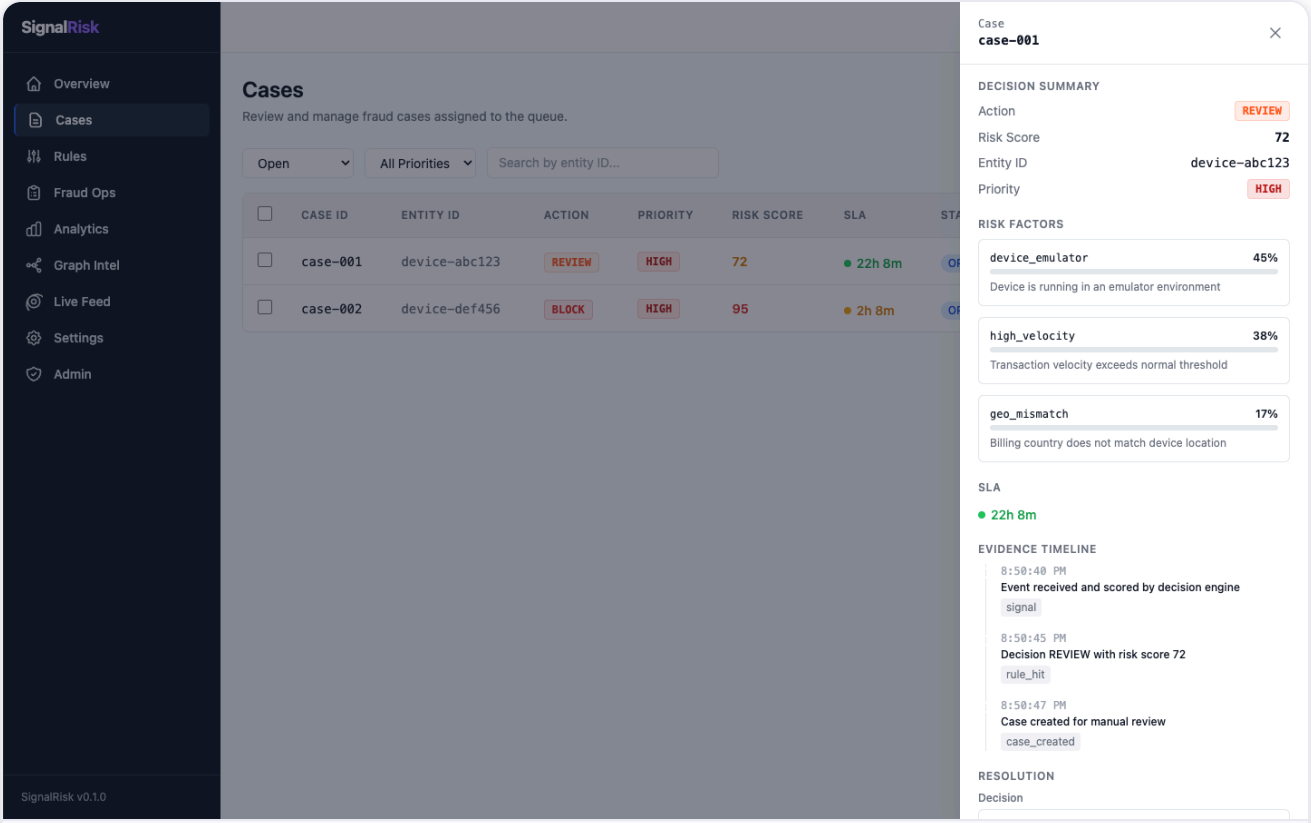
Understanding the Queue

Defaults to `OPEN` cases only. Cases automatically leave the queue when resolved or escalated. Use the status filter to view resolved or escalated cases.

Filtering

- **Status dropdown** — All / Open / In Review / Escalated / Resolved
- **Priority dropdown** — All / HIGH / MEDIUM / LOW
- **Search** — filter by Entity ID (300ms debounce)

Case Detail Panel



Case detail panel — decision summary, risk factor contribution bars, evidence timeline, and resolution form

Click **View** to open the sliding panel. It fetches the full case including evidence timeline.

Section	Description
Decision Summary	Automated decision: action, risk score, entity ID, priority
Risk Factors	Contributing signals with percentage bars and descriptions
SLA	Remaining time (turns red when critical)
Evidence Timeline	Ordered sequence: event received → scored → case created
Resolution	FRAUD / LEGITIMATE / INCONCLUSIVE + optional notes
Escalate	Send to senior analyst (OPEN cases only)

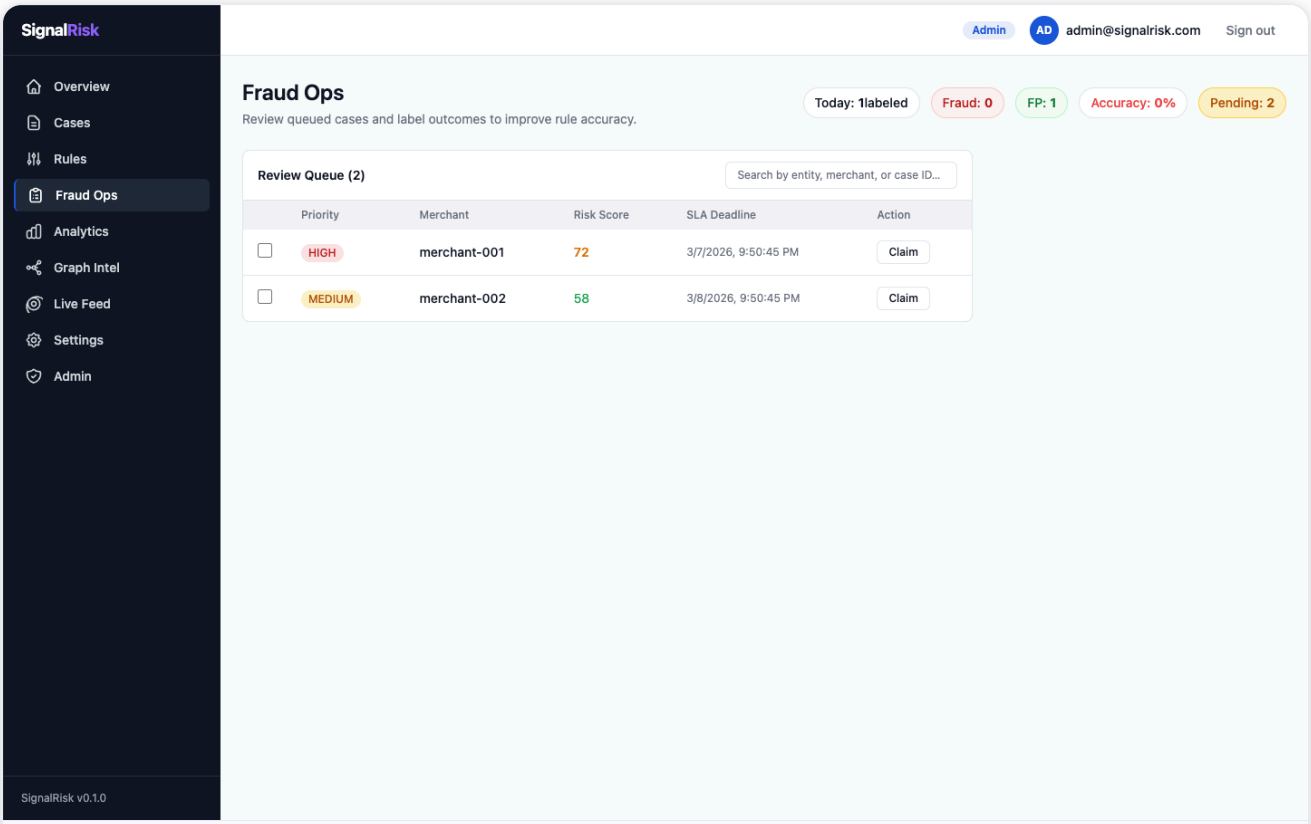
Resolving a Case

1. Click **View** on the row
2. Review risk factors and evidence timeline
3. Select **FRAUD**, **LEGITIMATE**, or **INCONCLUSIVE**
4. Optionally add notes
5. Click **Submit Resolution** — case immediately leaves the queue

> **TIP**
>
> Sort by SLA column to work the most urgent cases first. Cases with < 4h remaining are highlighted in red.

4 Fraud Ops

Case labeling to improve rule engine accuracy via automated feedback loop

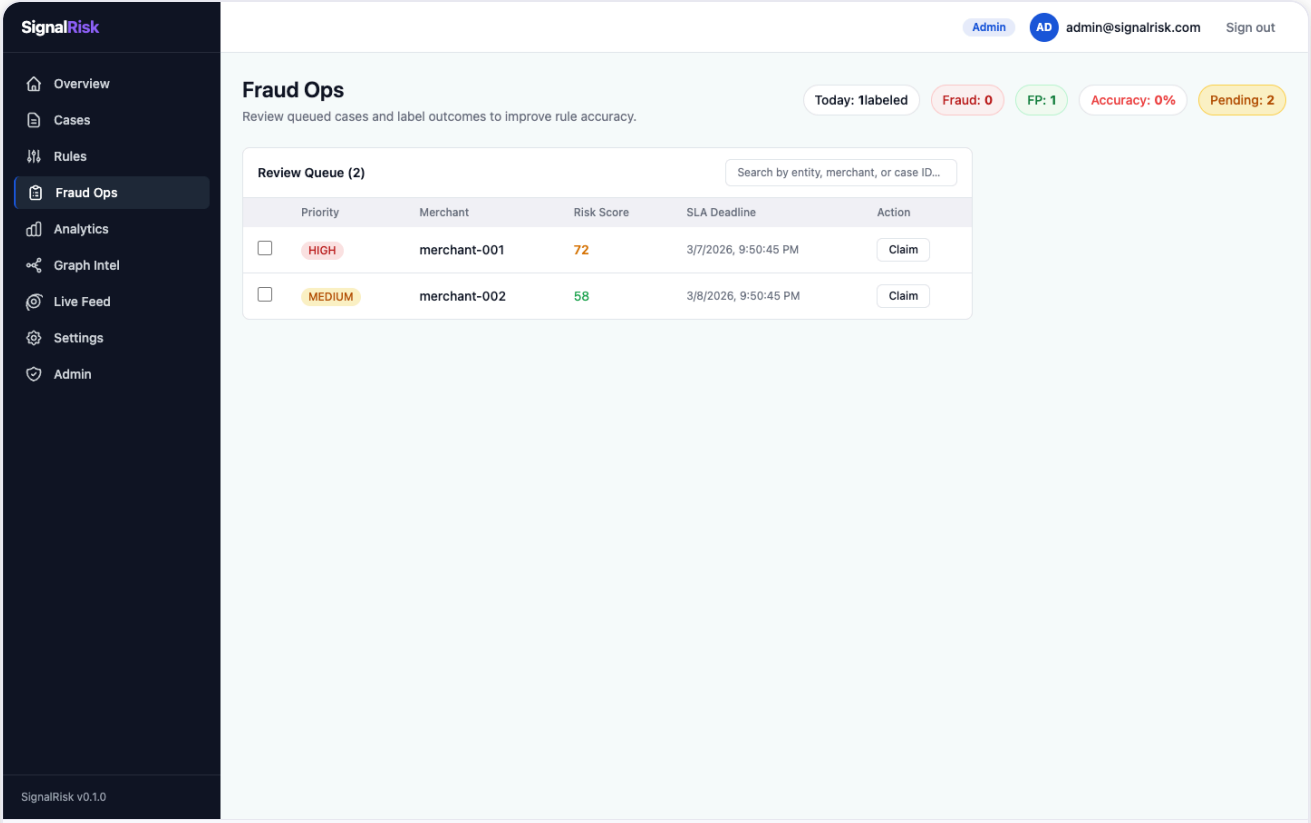


Fraud Ops page — labeling stats bar (top right), REVIEW-action queue with Claim and Label buttons

Labeling Stats Bar

Top-right stats update in real time: **Today: N labeled**, **Fraud: N**, **FP: N**, **Accuracy: N%**, **Pending: N**.

Outcome Modal



Label outcome modal — Fraud Confirmed, False Positive, or Inconclusive

Labeling a Case

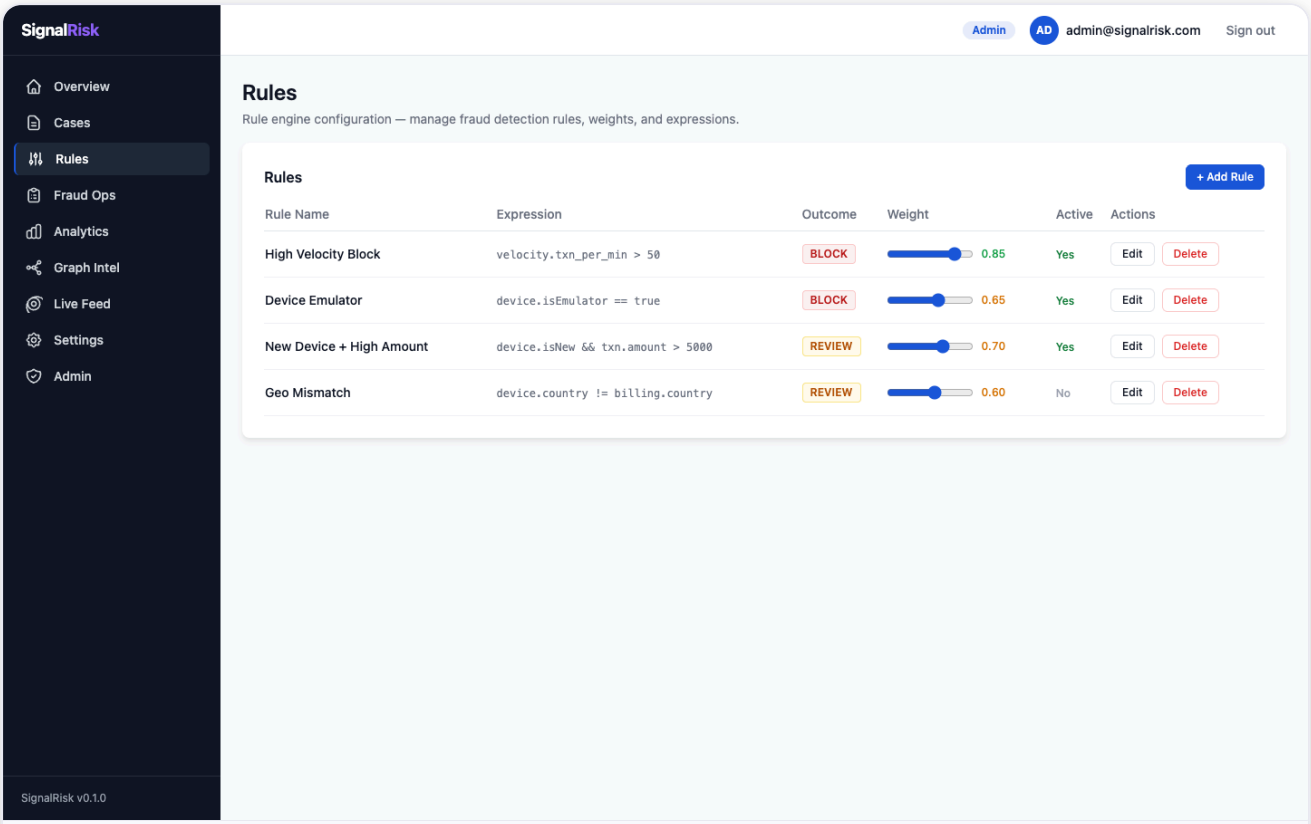
1. Click **Claim** to assign the case to yourself
2. Click **Label** to open the outcome modal
3. Select **Fraud Confirmed**, **False Positive**, or **Inconclusive**
4. Submit — case leaves queue and stats update immediately

Feedback Loop

Label	Rule weight effect
Fraud Confirmed	+0.05 (max 1.0) — rule correctly flagged fraud
False Positive	−0.03 (min 0.1) — rule over-triggered
Inconclusive	No change

5 Rules

DSL rule authoring, weight management, and active/inactive toggling

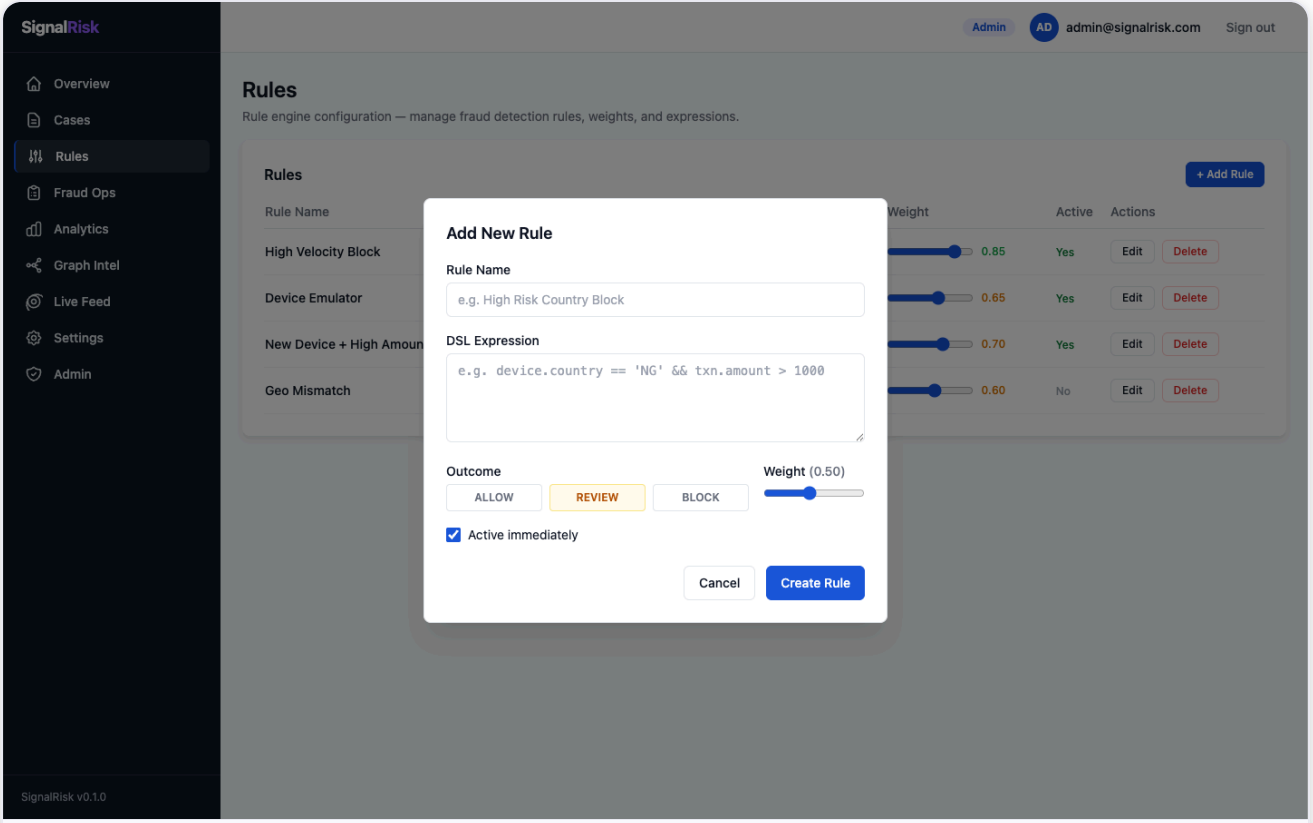


Rules page — rule list with DSL expressions, colour-coded outcomes, weight sliders, and toggle controls

Rules Table

Column	Description
Expression	DSL condition evaluated against each event (hover for full text)
Outcome	ALLOW REVIEW BLOCK
Weight	Contribution 0.10–1.00 — drag slider to adjust
Active	Toggle Yes/No — takes effect within seconds, no deployment needed

Adding a New Rule



Add Rule modal — name, DSL expression, outcome selector, weight slider, active toggle

Click **+ Add Rule**, fill in all fields, click **Create Rule**.

DSL Reference

Example expressions:

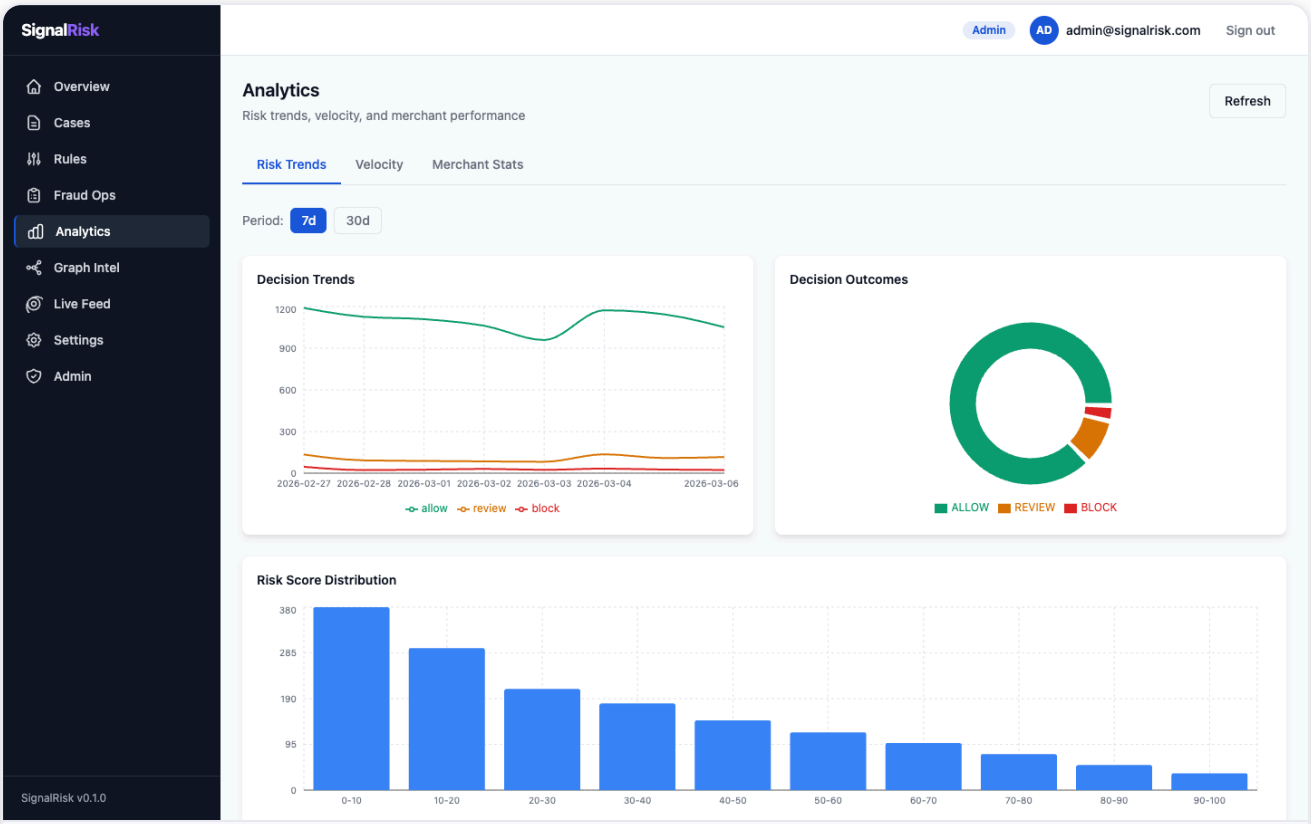
- `device.isEmulator == true && txn.amount > 100`
- `velocity.count_1h > 15`
- `network.vpnDetected == true && device.country == 'NG'`
- `telco.carrierRisk > 0.8 && txn.amount > 500`

Contexts: `device.` `velocity.` `network.` `telco.` `txn.`

> **WARNING**
>
> Deleting a rule is permanent. Consider deactivating first and monitoring for 24h before deleting.

6 Analytics

Historical performance metrics and merchant-level breakdowns



Analytics page — hourly event velocity chart and merchant performance breakdown table

Velocity Chart

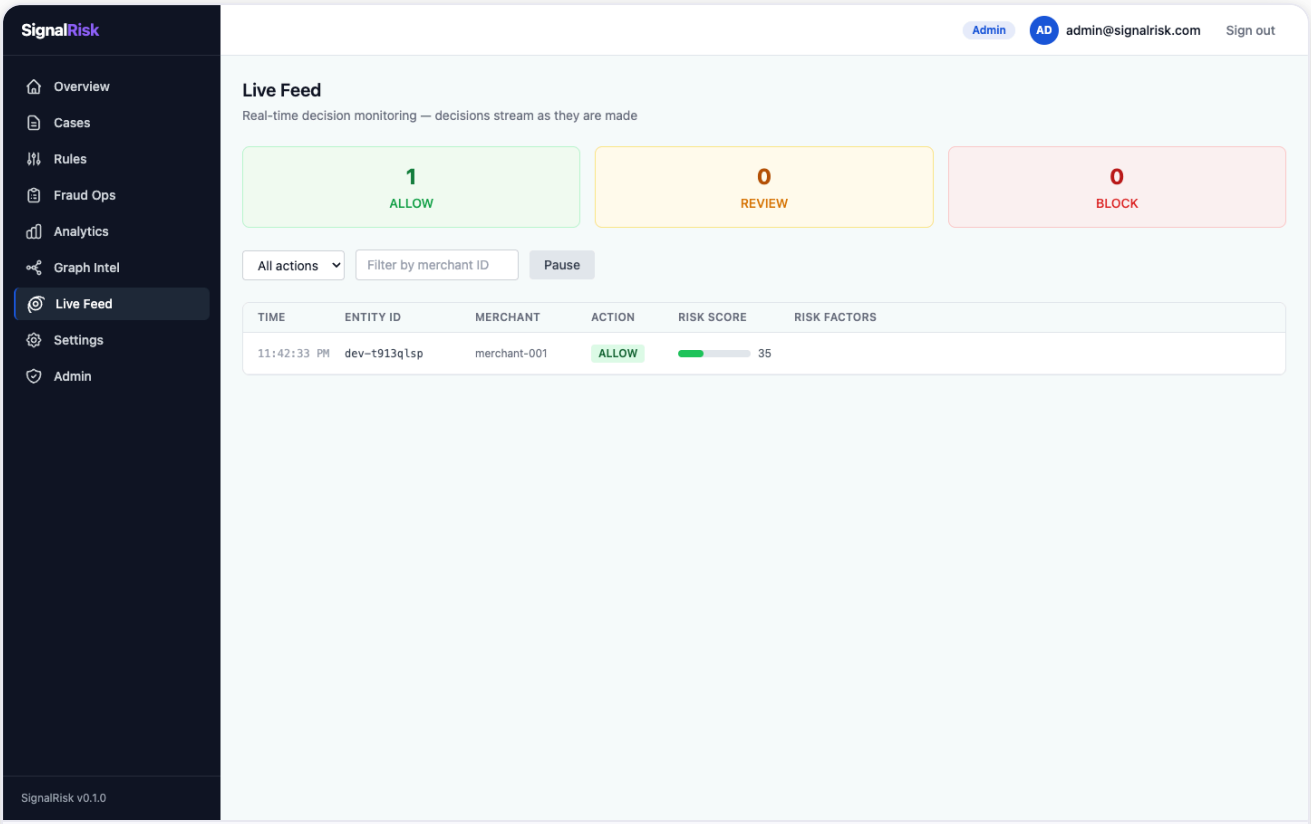
Time-series event volume per hour. Sudden BLOCK spikes without ALLOW increase → rule over-triggering or active attack.

Merchant Breakdown

Per-merchant: event volume, block rate, average risk score. Block rate > 15% or avg score > 60 warrant investigation.

7 Live Feed

Real-time full-screen decision stream for operations monitoring



Live Feed page — real-time decision stream with colour-coded badges and processing latency per event

Shows all decisions as they arrive via WebSocket. Retains the last 100 decisions and auto-scrolls. Display on a monitoring screen during incident response or rule testing.

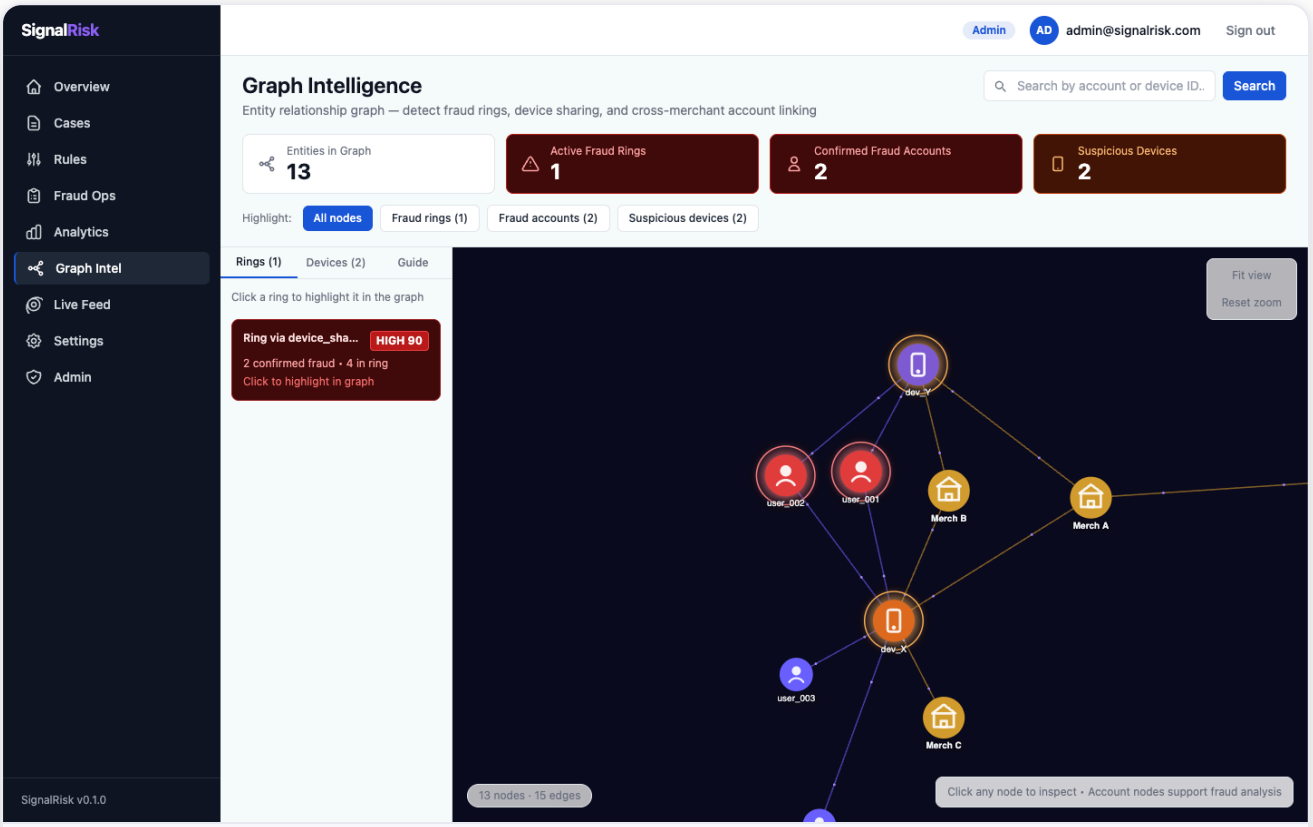
> **TIP**
>
> If the Connected indicator turns red, refresh the page to re-establish the WebSocket connection.

8 Graph Intelligence

Entity relationship graph — fraud ring detection, device sharing, and cross-merchant account linking

ACCESS

Available to Analyst and Admin roles. Requires Neo4j graph database connectivity.



Graph Intelligence page — force-directed entity graph with KPI summary, quick-filter bar, and left alert panel

What This Page Shows

The graph visualizes relationships between **accounts**, **devices**, and **merchants** detected from processed events. It surfaces three types of threats:

Threat Type	Description
Fraud Ring	Multiple accounts sharing the same device, with at least one confirmed fraud account in the cluster
Suspicious Device	A single device appearing across multiple merchant accounts — a cross-merchant sharing signal
Emulator	Device identified as a software emulator (not a real physical handset)

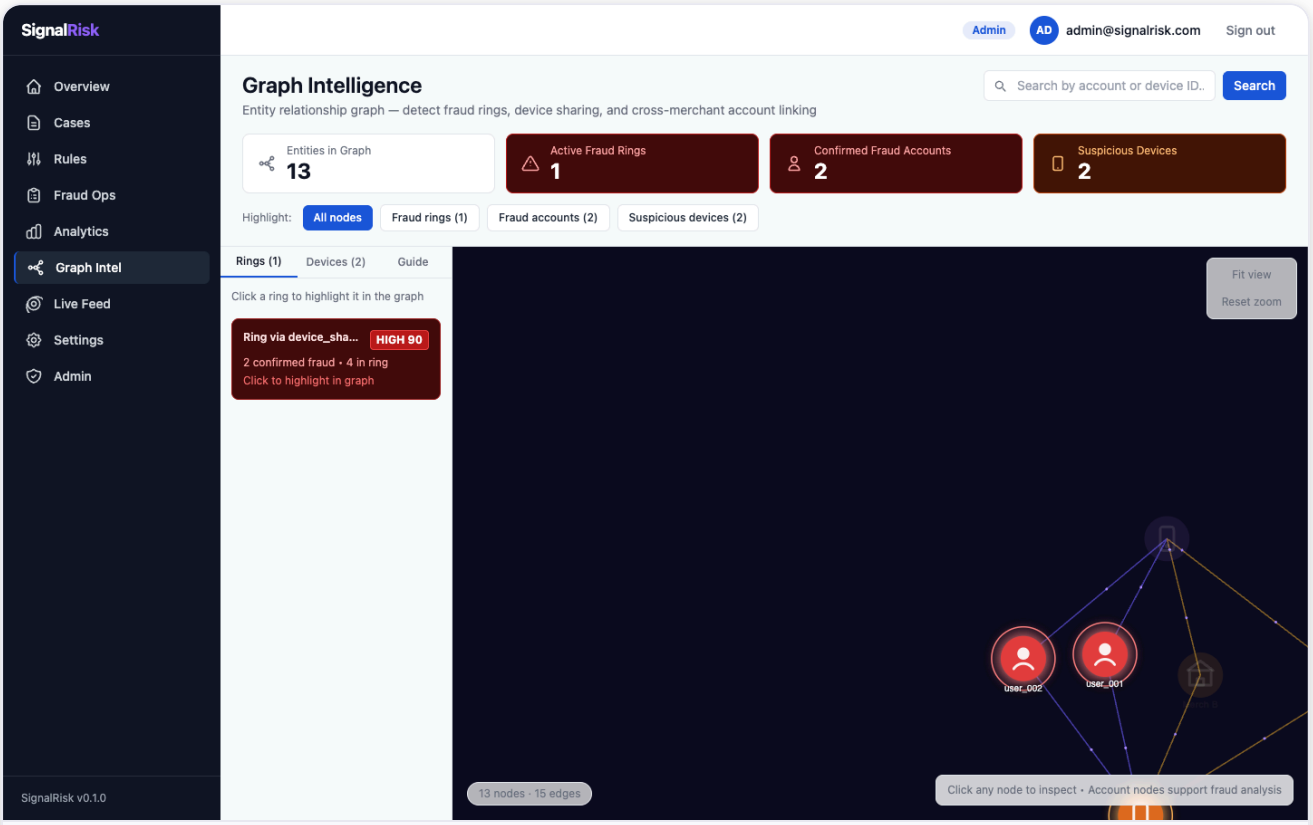
Reading the Graph

Node Shape	Type	Color

Circle with person icon	Account	Indigo — red if fraud confirmed
Circle with phone icon	Device	Green — orange if suspicious, purple if emulator
Circle with store icon	Merchant	Amber

High-risk nodes (fraud / suspicious) display a colored glow ring. Hover any node to see its ID tooltip.

Fraud Ring Alerts



Fraud ring highlighted — clicking a ring card in the left panel zooms to and highlights all member nodes

The **Rings** tab in the left panel lists every detected fraud ring with risk score and member counts. Clicking a ring card highlights all its nodes and centers the camera on them.

Node Inspection & Fraud Analysis

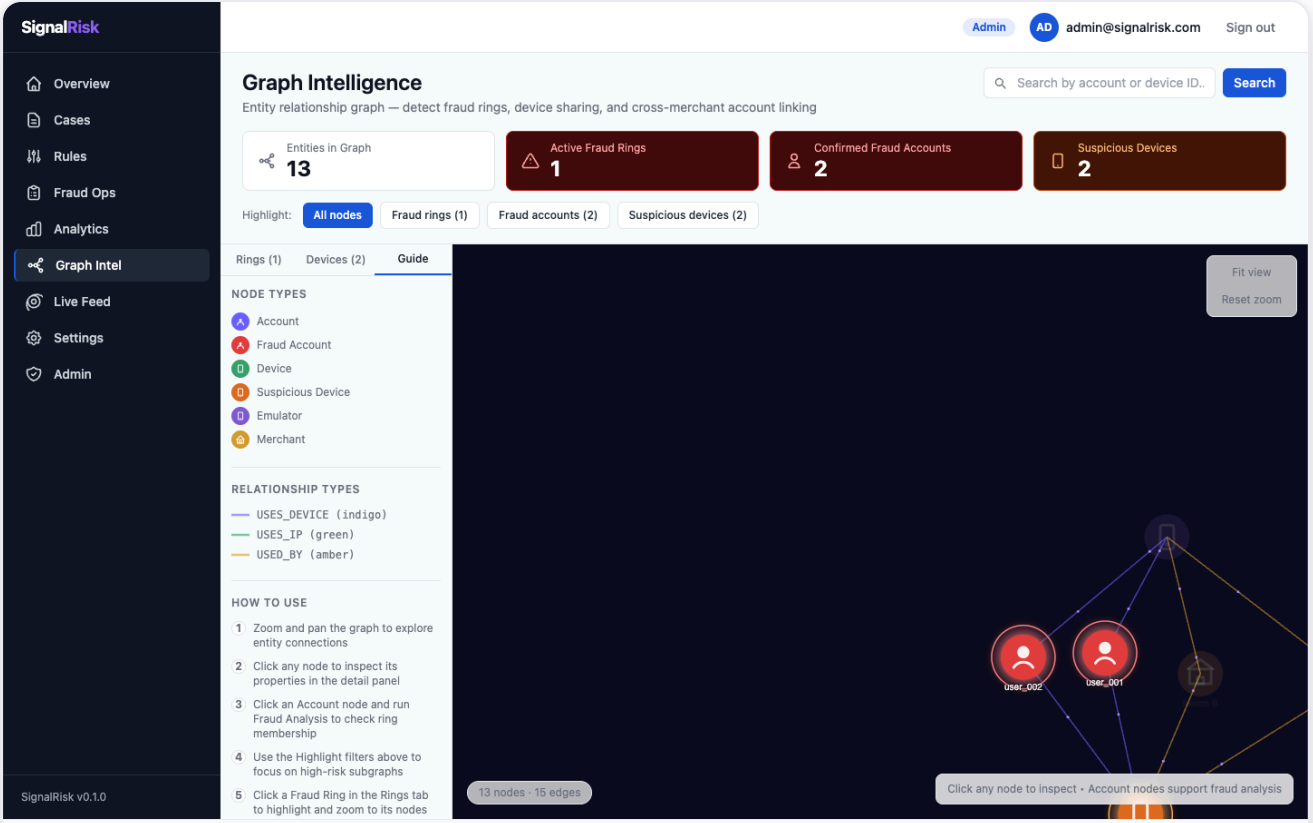
Click any node to open the detail panel on the right. For Account nodes, click **Run Fraud Analysis** to query the graph for:

- Risk score (0–100)
- Number of fraud neighbors (directly connected fraud accounts)
- Number of accounts sharing a device
- Fraud ring membership

Quick Filters

Use the **Highlight** filter bar to instantly focus on a subgraph: Fraud rings, Fraud accounts, or Suspicious devices. Non-matching nodes fade to 12% opacity.

Guide Tab



Guide tab — node type legend with icons, relationship type color key, and step-by-step usage instructions

TIP

Use

FIT VIEW

(top-right canvas control) to reset the camera after zooming into a fraud ring.

RESET ZOOM

returns to 1:1 scale.

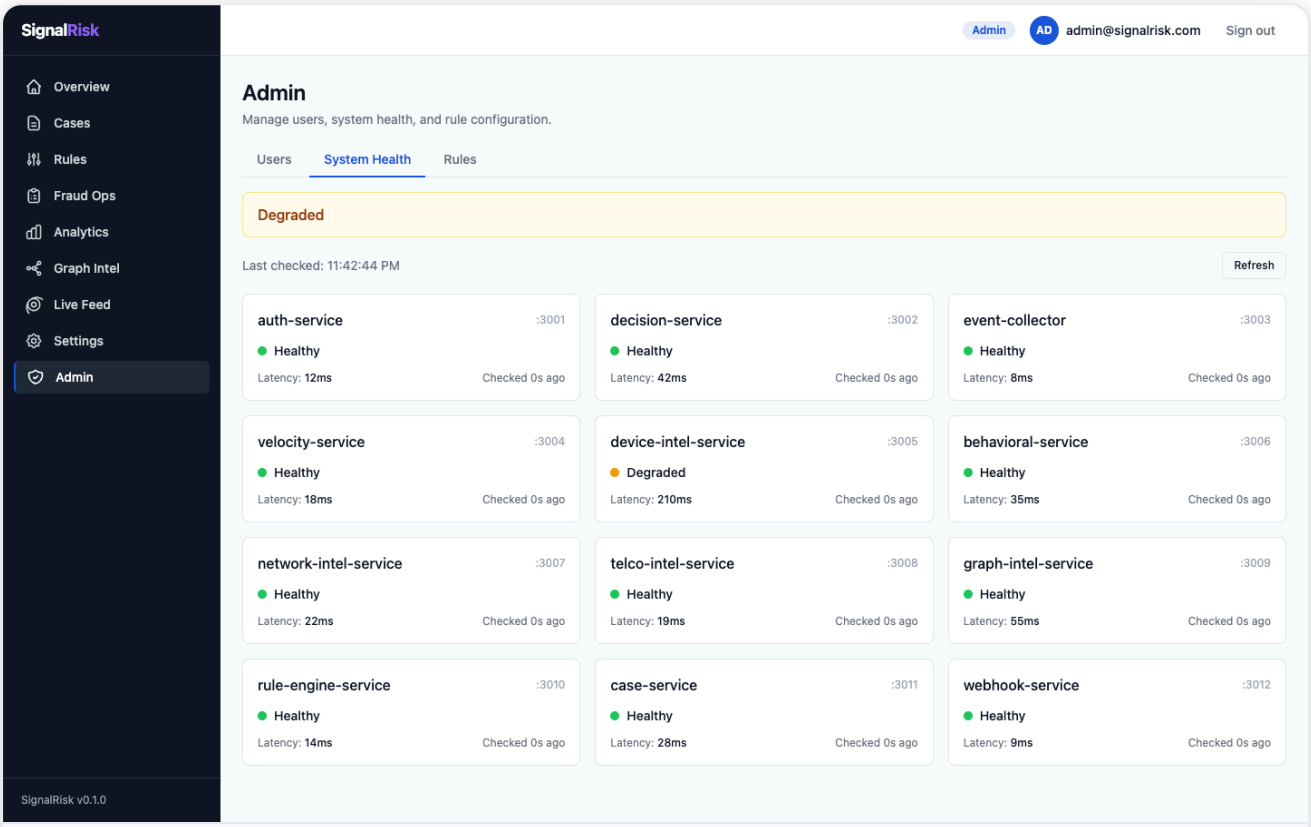
9 Admin

System health monitoring, user management, and rule administration

ACCESS RESTRICTED

Only visible to Admin role accounts.

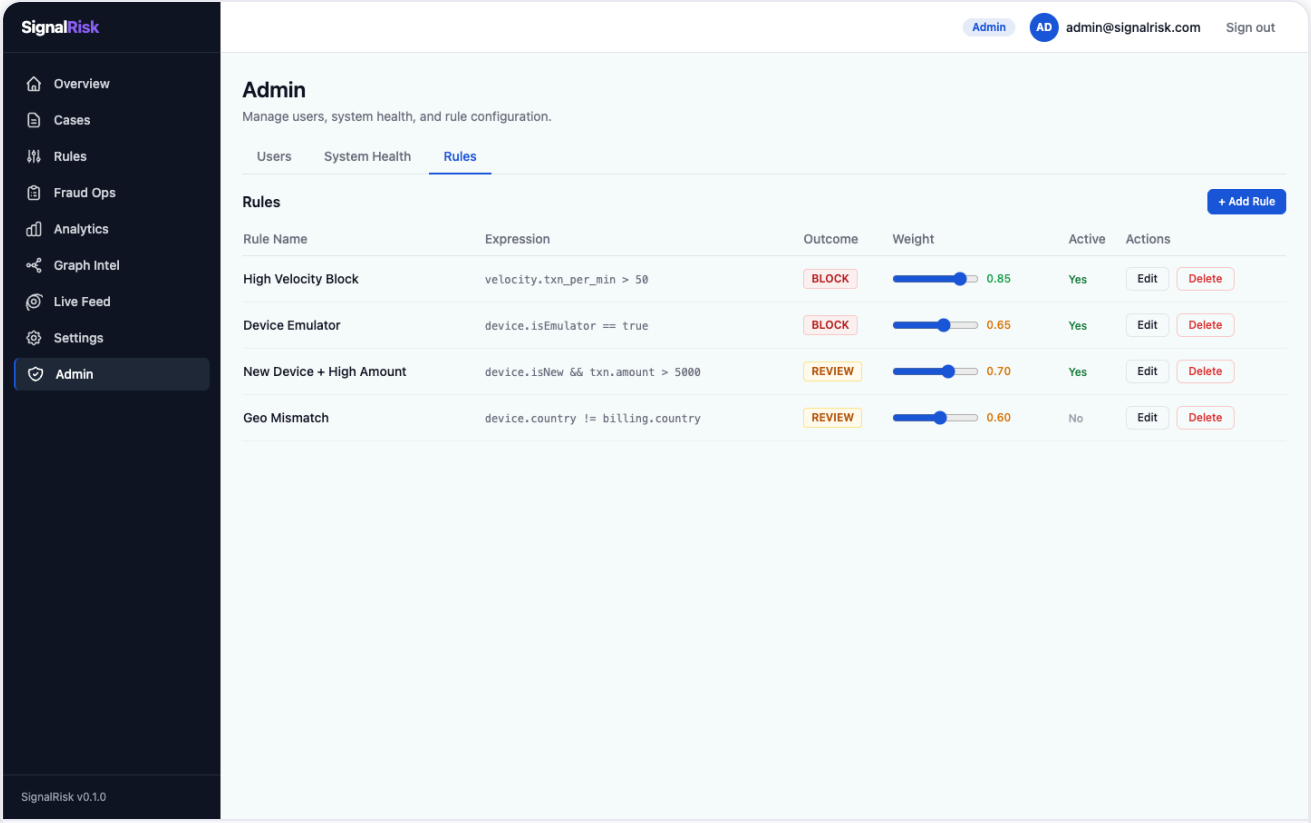
System Health Tab



Admin — System Health tab showing real-time status of all 13 microservices with response times

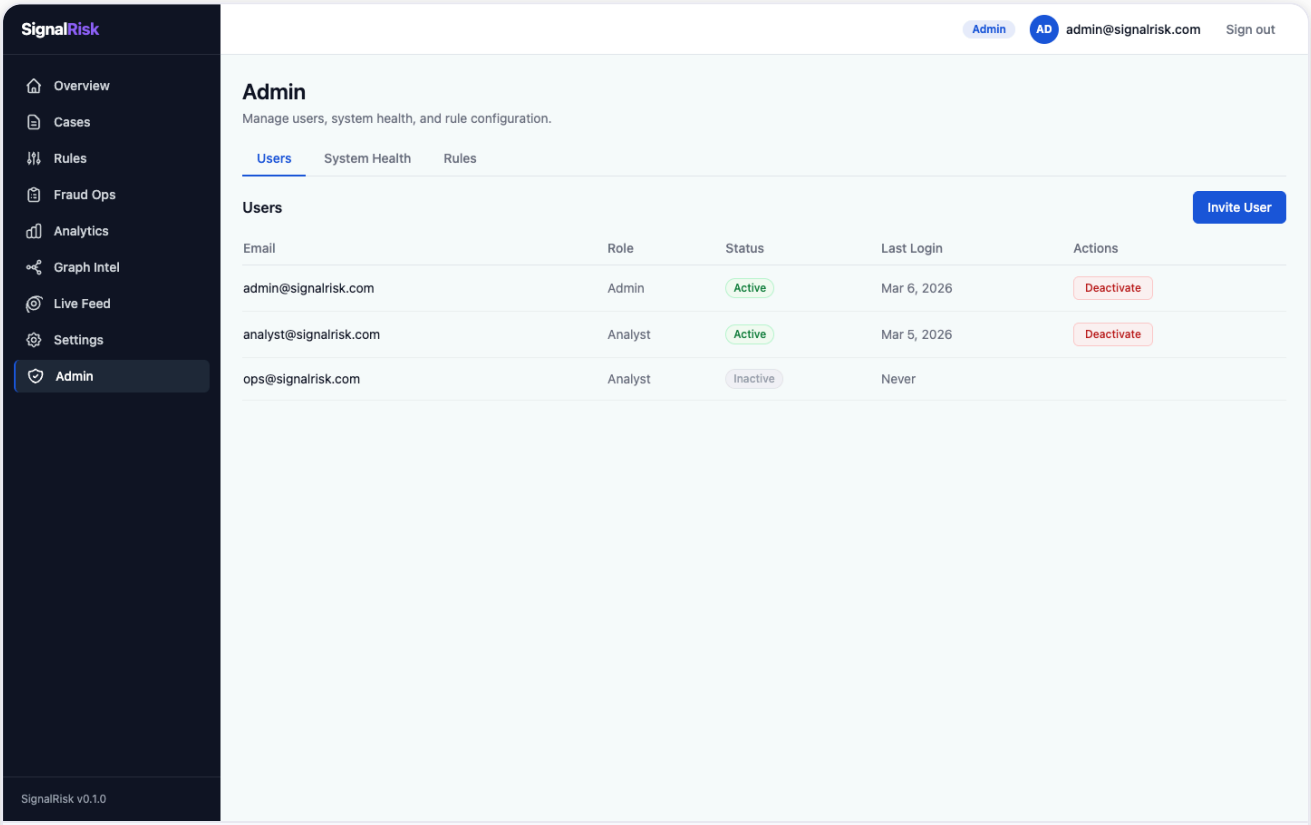
Each service card: status indicator (green/amber/red), response time, port, last checked timestamp.

Rules Tab



Admin — Rules tab with full rule CRUD (identical to the standalone Rules page)

Users Tab

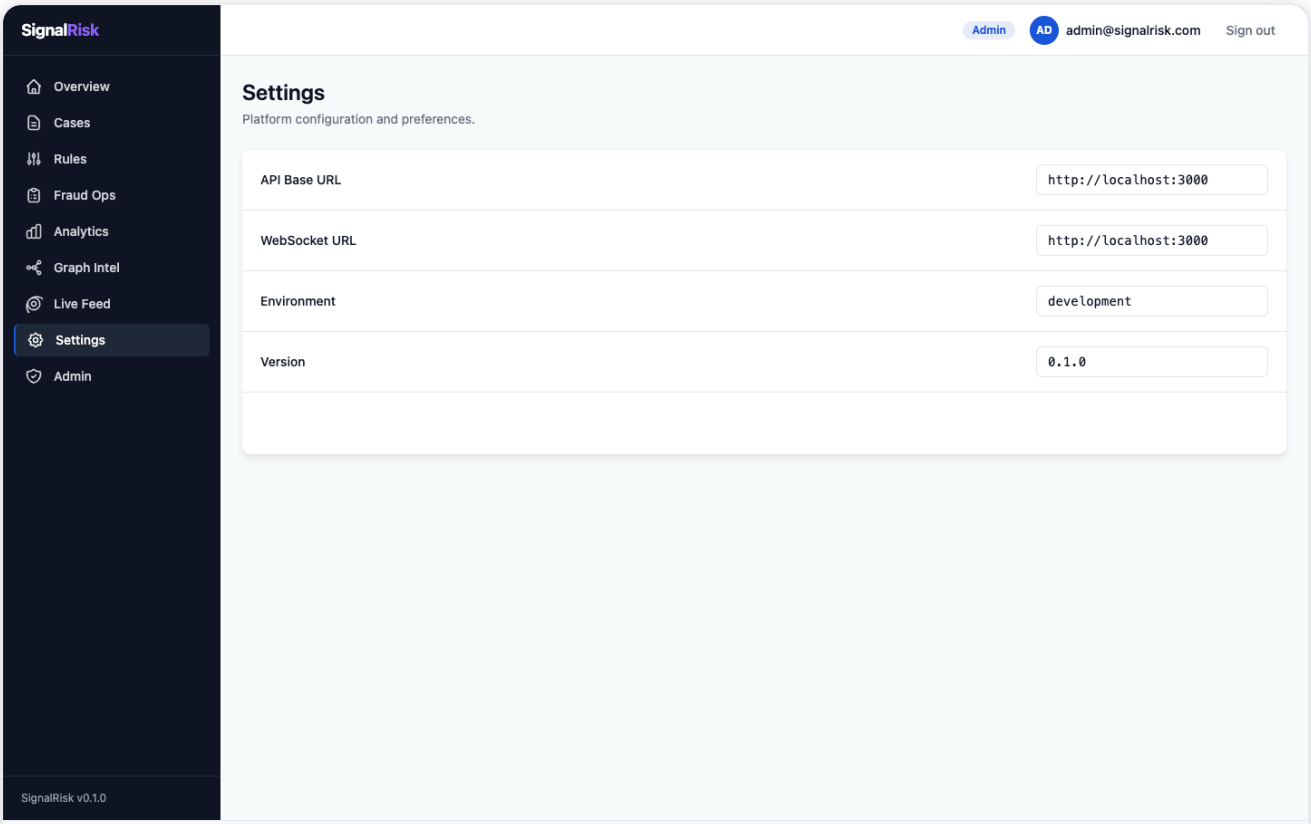


Admin — Users tab listing all analyst and admin accounts with roles and last login

Use to onboard analysts, change roles, or deactivate accounts.

10 Settings

Platform configuration and preferences



Settings page — editable API URL, WebSocket URL, environment, and version fields

Setting	Default	Description
API Base URL	`http://localhost:3000`	Base URL for all API calls
WebSocket URL	`http://localhost:3000`	WebSocket server for live stream
Environment	`development`	Environment label in UI
Version	`0.1.0`	Platform version number

Edit any field and click **Save Settings**. Settings are stored in `localStorage` and survive page reloads.

11 Roles & Permissions

What each role can access and modify

Feature	Admin	Analyst	Merchant
Overview	Read	Read	Own data
Cases — view	All	All	Own only
Cases — resolve/escalate	Yes	Yes	No
Fraud Ops — label	Yes	Yes	No
Rules — view	Yes	Yes	No
Rules — create/edit/delete	Yes	No	No
Analytics	All	All	Own only
Live Feed	Yes	Yes	No
Graph Intelligence	Yes	Yes	No
Admin panel	Yes	No	No
Settings	Yes	Yes	Yes

12 Tips, Best Practices & Shortcuts

Productivity tips for daily operations

Case Queue

- Press **Esc** or click backdrop to close the detail panel
- Search by Entity ID — 300ms debounce, type quickly
- After resolving/escalating, queue refreshes automatically
- Header checkbox selects all visible cases for bulk action

SLA Management

- BLOCK = 4h SLA, REVIEW = 24h SLA
- Red SLA = breach imminent — escalate if you can't resolve now
- Sort SLA column ascending for urgency order

Rule Authoring

- Start at **weight 0.5** — let the feedback loop tune it
- Deactivate before deleting — preserves history
- Combine signals: `device.isEmulator && txn.amount > 500`
- Watch Live Feed after activating a new rule for unexpected BLOCK spikes

Fraud Ops

- Batch label obvious fraud clusters for efficiency
- Target 80%+ daily accuracy for meaningful weight adjustments

INCIDENT RESPONSE CHECKLIST

1. Check Overview KPIs 2. Admin → System Health for degraded services 3. Live Feed for decision pattern anomalies 4. Analytics for traffic spike origin 5. Rules page to adjust or deactivate rule if needed